

Weak forms and problem setup — Two-segment Timoshenko cantilever beam vibration eigenproblem

Problem setup

The computational domain is the beam axis

$$\Omega = (0, L),$$

split into two subdomains

$$\Omega_1 = (0, x_s), \quad \Omega_2 = (x_s, L),$$

where x_s is the interface location between the two beam segments.

The unknown fields are

$$w(x) \quad (\text{transverse displacement}), \quad \theta(x) \quad (\text{cross-section rotation}).$$

The beam is clamped at the left end, so the essential boundary conditions are

$$w(0) = 0, \quad \theta(0) = 0.$$

The right end $x = L$ is free.

Schematic

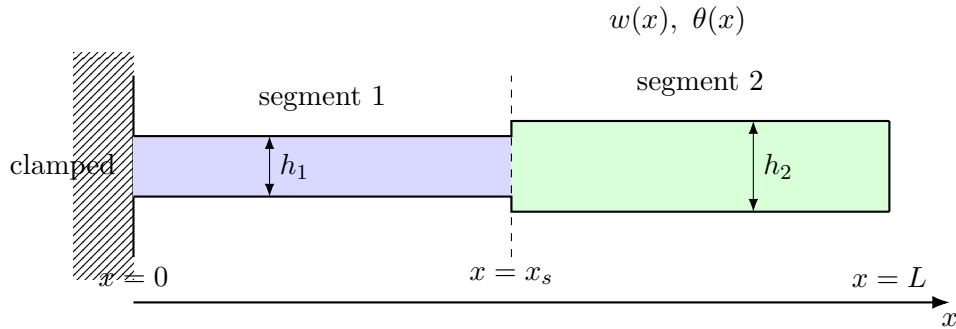


Figure 1: Two-segment cantilever beam with piecewise section thickness.

Function spaces

Define

$$V := \{v \in H^1(\Omega) : v(0) = 0\}.$$

Then the trial and test fields satisfy

$$(w, \theta) \in V \times V, \quad (v, \phi) \in V \times V.$$

Weak formulation

We seek the eigenpair

$$((w, \theta), \lambda) \in (V \times V) \times \mathbb{R}$$

such that

$$a((w, \theta), (v, \phi)) = \lambda m((w, \theta), (v, \phi)) \quad \forall (v, \phi) \in V \times V.$$

The bilinear form for stiffness is

$$a((w, \theta), (v, \phi)) = \sum_{r=1}^2 \int_{\Omega_r} [EI_r \theta'(x) \phi'(x) + kGA_r (w'(x) - \theta(x)) (v'(x) - \phi(x))] dx.$$

The mass bilinear form is

$$m((w, \theta), (v, \phi)) = \sum_{r=1}^2 \int_{\Omega_r} [\rho A_r w(x) v(x) + \rho I_r \theta(x) \phi(x)] dx.$$

Expanding the shear term gives the equivalent expression

$$\sum_{r=1}^2 \int_{\Omega_r} [kGA_r w' v' - kGA_r \theta v' - kGA_r w' \phi + EI_r \theta' \phi' + kGA_r \theta \phi] dx = \lambda \sum_{r=1}^2 \int_{\Omega_r} [\rho A_r w v + \rho I_r \theta \phi] dx.$$

Section coefficients

For each segment $r \in \{1, 2\}$, let the rectangular cross-section have width b and thickness h_r . Then

$$A_r = b h_r, \quad I_r = \frac{b h_r^3}{12}, \quad G = \frac{E}{2(1 + \nu)}.$$

Hence the coefficients are

$$EI_r = EI_r, \quad kGA_r = \kappa GA_r, \quad \rho A_r = \rho A_r, \quad \rho I_r = \rho I_r.$$

Remarks

This is a free-vibration generalized eigenvalue problem, so no external load functional appears in the weak statement.

The heterogeneity enters through the piecewise segment thicknesses h_1 and h_2 , which induce different values of EI_r , kGA_r , ρA_r , and ρI_r in the two subdomains.